

LAB CYCLE 1

Create a table 'STUDENT' with the following details including marks of 3 subjects ('english', 'maths', 'IT' and the given constraints:

COLUMN NAME	DATA TYPE	CONSTRAINTS
sid	int	PRIMARY KEY
name	varchar(15)	NOT NULL, UNIQUE
course	varchar(3)	'BCA', 'BBA','BA ENGLISH'
english	int	NOT NULL, <=50
maths	int	NOT NULL,<=50
IT	int	NOT NULL,<=50
college	varchar(15)	SGC ARUVITHURA

```
mysql>create table student(sid int primary key,name varchar(20) not null unique,course varchar(20)
check(course in('bca','bba','ba english')),english int not null check(english<=50),maths int not null
check(maths<=50),it int not null check(it<=50),college varchar(20)default 'sgc aruvithura');
```

Write SQL queries for the following questions based on above table:

1. Describe the table 'STUDENT'.

```
mysql>desc student;
```

Field	Type	Null	Key	Default	Extra
sid	int	NO	PRI	NULL	
name	varchar(20)	NO	UNI	NULL	
course	varchar(20)	YES		NULL	
english	int	NO		NULL	
maths	int	NO		NULL	
it	int	NO		NULL	
college	varchar(20)	YES		sgc aruvithura	

2. Insert 8 records into the table.

```
mysql>insert into student values(1,'benson','bba',45,36,43,'sgc aruvithura');
mysql>insert into student values(2,'amal','bba',42,30,41,'sgc aruvithura');
mysql>insert into student values(3,'josna','bca',41,34,33,'sgc aruvithura');
mysql>insert into student values(4,'karun','bca',24,34,44,'sgc aruvithura');
mysql>insert into student values(5,'nihil','bca',25,44,24,'sgc aruvithura');
mysql>insert into student values(6,'justin','ba english',40,45,48,'sgc aruvithura');
mysql>insert into student values(7,'aravind','ba english',49,29,43,'sgc aruvithura');
mysql>insert into student values(8,'gokul','bba',47,25,42,'sgc aruvithura');
```

3. Display the details of all students.

```
mysql>select * from student;
```

sid	name	course	english	maths	it	college
1	benson	bba	45	36	43	sgc aruvithura
2	amal	bba	42	30	41	sgc aruvithura
3	josna	bca	41	34	33	sgc aruvithura
4	karun	bca	24	34	44	sgc aruvithura
5	nihil	bca	25	44	24	sgc aruvithura
6	justin	ba english	40	45	48	sgc aruvithura
7	aravind	ba english	49	29	43	sgc aruvithura
8	gokul	bba	47	25	42	sgc aruvithura

4. Insert a new field 'total' and update it with total marks of each student.

```
mysql>alter table student add total int;
```

```
mysql>update student set total=english+maths+it;
```

```
mysql>select * from student;
```

sid	name	course	english	maths	it	college	total
1	benson	bba	45	36	43	sgc aruvithura	124
2	amal	bba	42	30	41	sgc aruvithura	113
3	josna	bca	41	34	33	sgc aruvithura	108
4	karun	bca	24	34	44	sgc aruvithura	102
5	nihil	bca	25	44	24	sgc aruvithura	93
6	justin	ba english	40	45	48	sgc aruvithura	133
7	aravind	ba english	49	29	43	sgc aruvithura	121
8	gokul	bba	47	25	42	sgc aruvithura	114

5. Display the details of all students who have joined for the course 'BCA'.

```
mysql>select * from student where course='bca';
```

sid	name	course	english	maths	it	college	total
3	josna	bca	41	34	33	sgc aruvithura	108
4	karun	bca	24	34	44	sgc aruvithura	102
5	nihil	bca	25	44	24	sgc aruvithura	93

6. Find all students whose names start with 'B'.

```
mysql>select name from student where name like 'b%';
```

```
+-----+
| name  |
+-----+
| benson |
+-----+
```

7. Display the details of students in the sorted order of their total marks.

```
mysql>select * from student order by total asc;
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
| sid | name  | course | english | maths | it | college          | total |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 5 | nihil | bca    | 25 | 44 | 24 | sgc aruvithura | 93 |
| 4 | karun | bca    | 24 | 34 | 44 | sgc aruvithura | 102 |
| 3 | josna | bca    | 41 | 34 | 33 | sgc aruvithura | 108 |
| 2 | amal  | bba    | 42 | 30 | 41 | sgc aruvithura | 113 |
| 8 | gokul | bba    | 47 | 25 | 42 | sgc aruvithura | 114 |
| 7 | aravind | ba english | 49 | 29 | 43 | sgc aruvithura | 121 |
| 1 | benson | bba    | 45 | 36 | 43 | sgc aruvithura | 124 |
| 6 | justin | ba english | 40 | 45 | 48 | sgc aruvithura | 133 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

8. Display the course name as 'course' and count of students as 'count' joined for each course.

```
mysql>select course as course_name,count(*)as count from student group by course;
```

```
+-----+-----+
| course_name | count |
+-----+-----+
| bba        | 3 |
| bca        | 3 |
| ba english | 2 |
+-----+-----+
```

9. Find all students whose names ends with 'A' and joined for the course 'BCA'.

```
mysql>select * from student where name like '%a'and course='bca';
```

```
+-----+-----+-----+-----+-----+-----+-----+-----+
| sid | name  | course | english | maths | it | college          | total |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 3 | josna | bca    | 41 | 34 | 33 | sgc aruvithura | 108 |
+-----+-----+-----+-----+-----+-----+-----+-----+
```

10. Find the highest marks scored by student of 'BCA' for the subject 'english'.

mysql>select max(english)from student group by course having course='bca';

```
+-----+
| max(english) |
+-----+
|           41 |
+-----+
```

11. Delete the details of students whose name contains 'A' as the second character.

mysql>delete from student where name like '_a%';

12. Display the course name and highest mark scored for each subject in each course except 'BCA'.

mysql>select course, max(english), max(maths), max(it) from student group by course having course !='bca';

```
+-----+-----+-----+-----+
| course      | max(english) | max(maths) | max(it) |
+-----+-----+-----+-----+
| bba         |           47 |           36 |         43 |
| ba english  |           49 |           45 |         48 |
+-----+-----+-----+-----+
```

13. Create a new table 'STUDENT_DETAILS' from the table 'STUDENT'.

mysql>create table student_details as select * from student;

14. Delete all records from the table 'STUDENT'.

mysql> delete from student;

15. Delete the table 'STUDENT'.

mysql>drop table student;